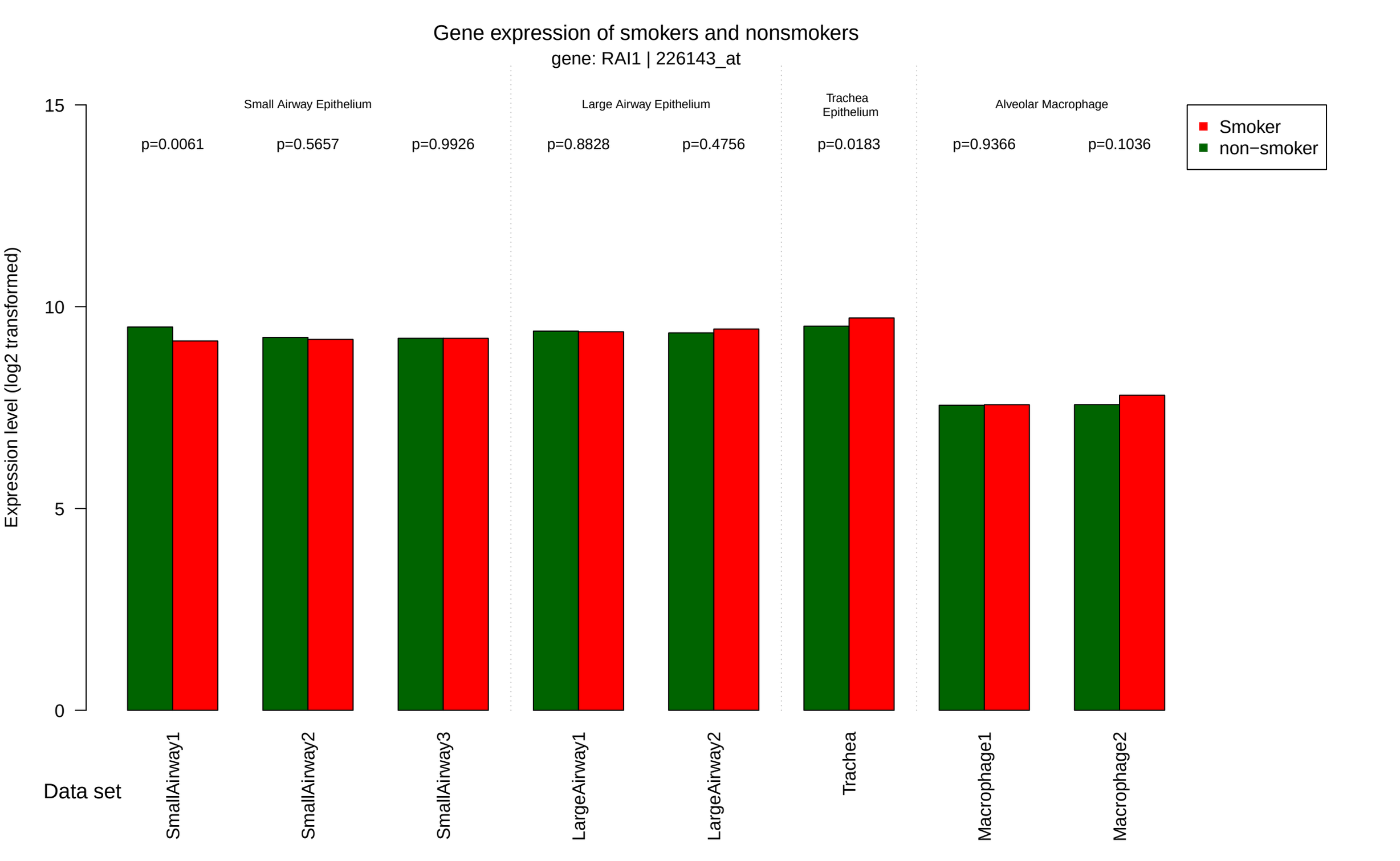
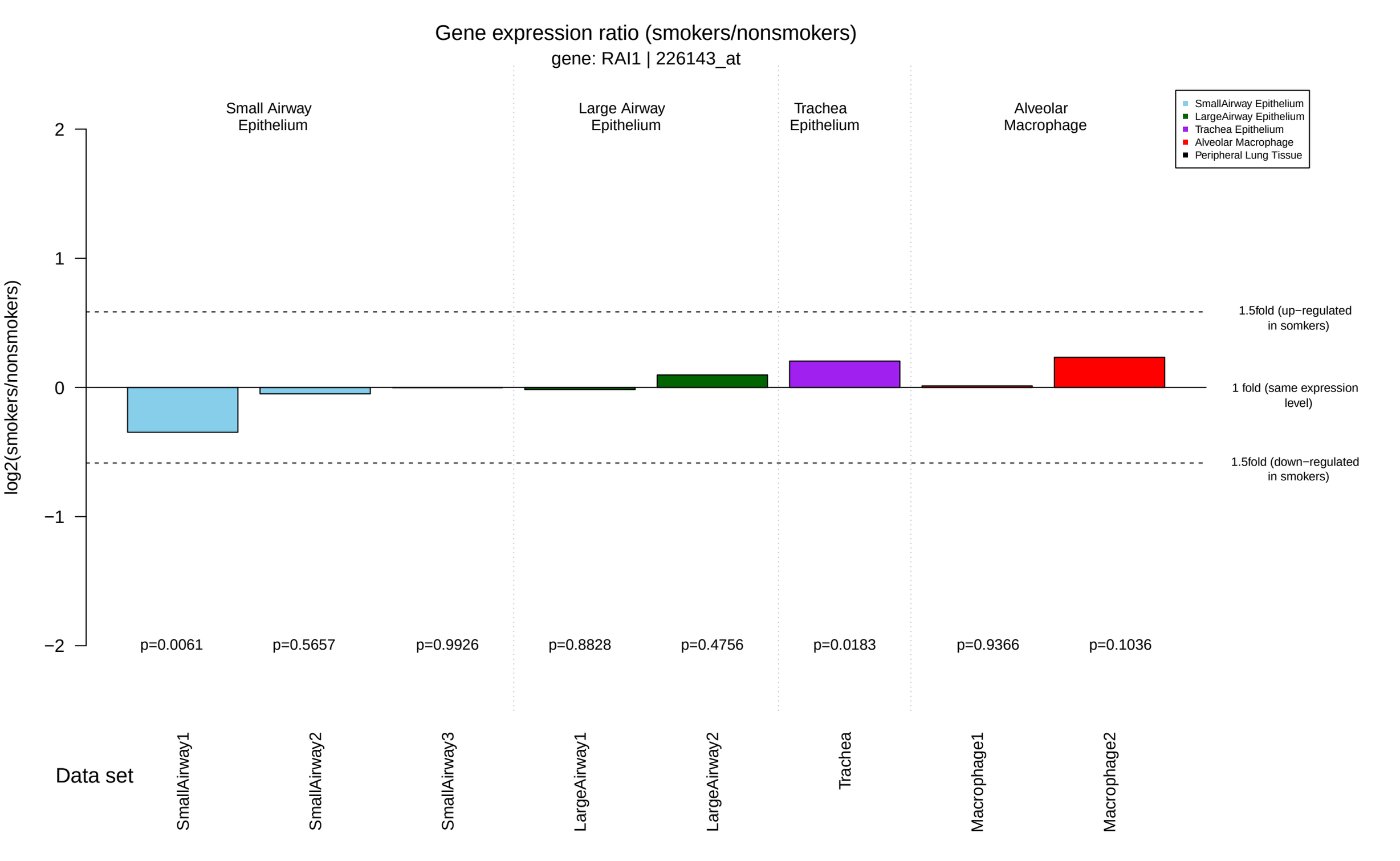
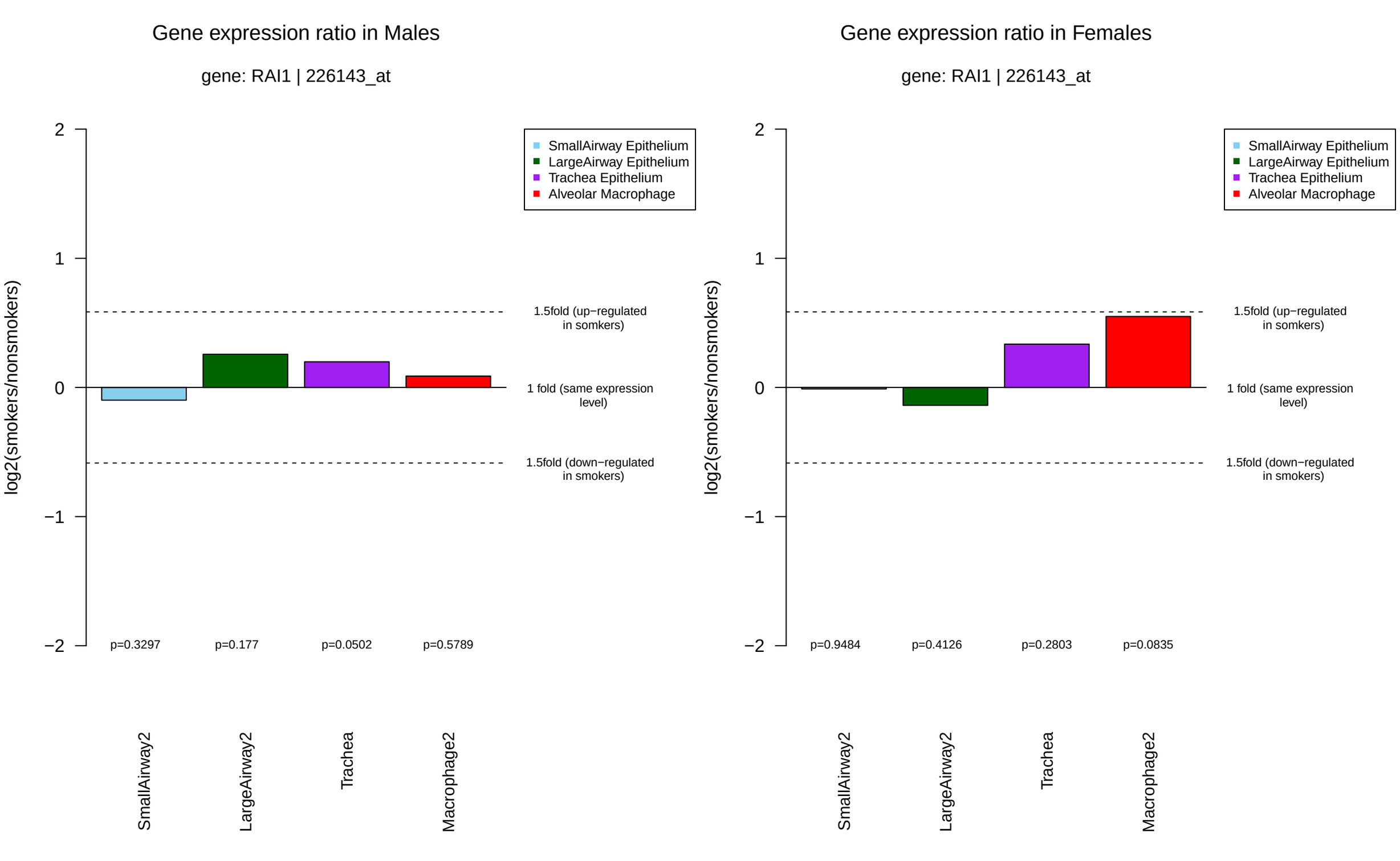
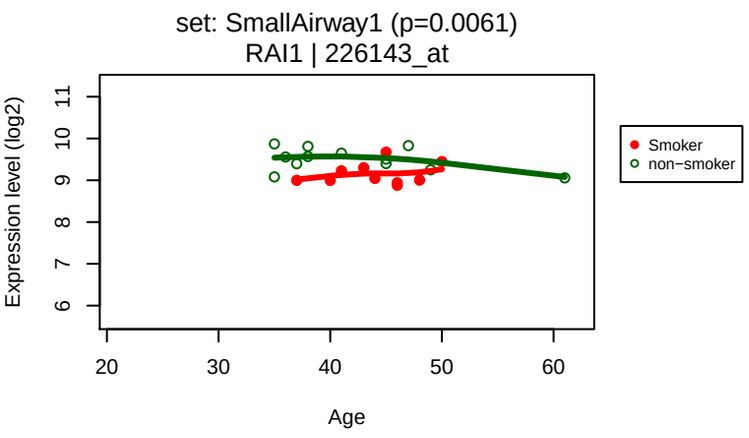
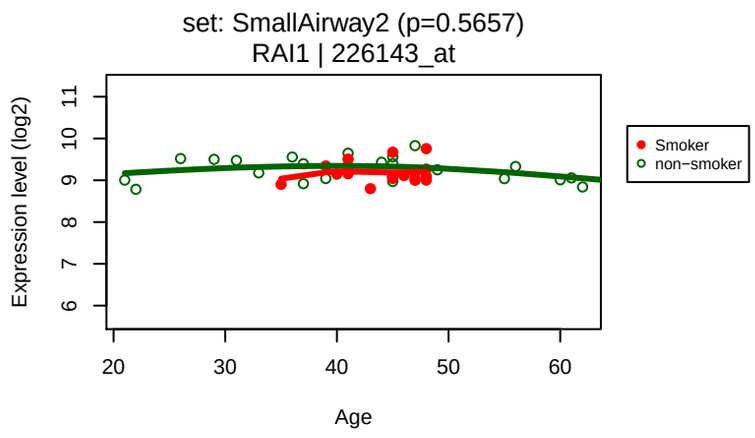
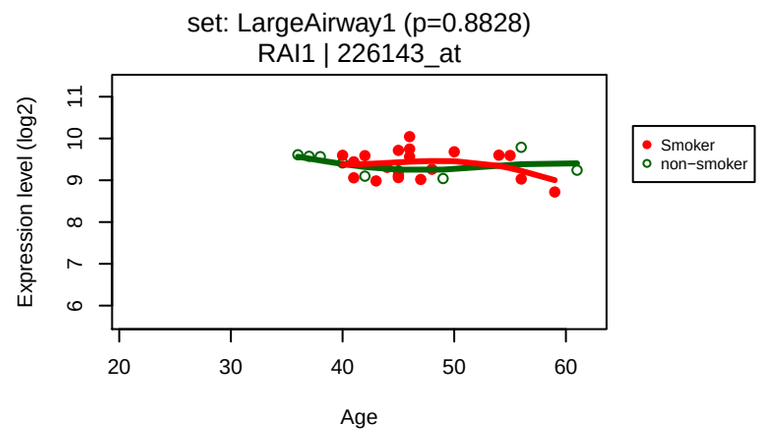

Gene expression of smokers and nonsmokers
gene: RAI1 | 226143_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0061
p=0.5657
p=0.9926
p=0.8828
p=0.4756
p=0.0183
p=0.9366
p=0.1036
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

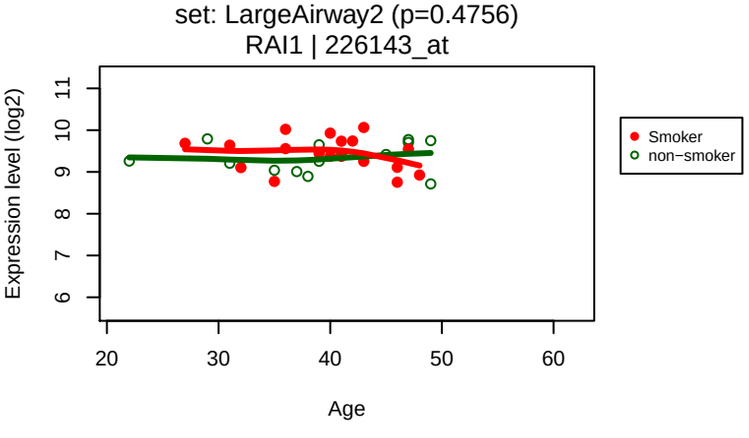

Gene expression ratio (smokers/nonsmokers)
gene: RAI1 | 226143_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0061
p=0.5657
p=0.9926
p=0.8828
p=0.4756
p=0.0183
p=0.9366
p=0.1036
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

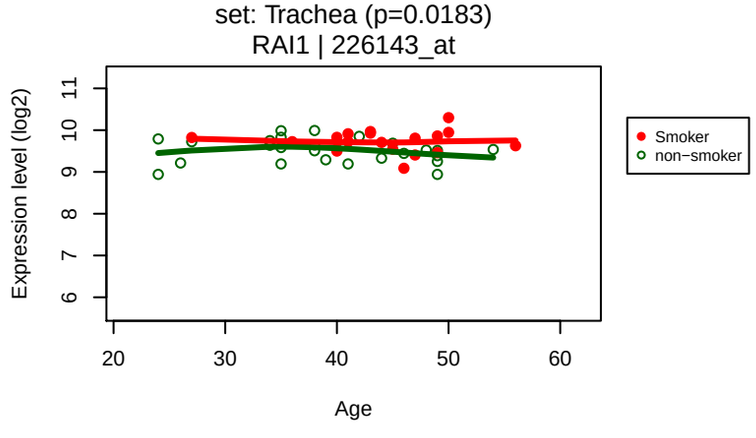

Gene expression ratio in Males
gene: RAI1 | 226143_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3297
p=0.177
p=0.0502
p=0.5789
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAI1 | 226143_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9484
p=0.4126
p=0.2803
p=0.0835
SmallAirway2
LargeAirway2
Trachea
Macrophage2

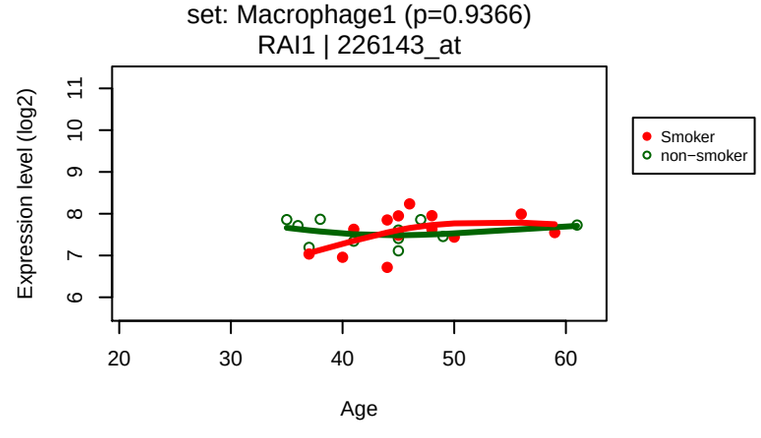

set: SmallAirway1 (p=0.0061)
RAI1 | 226143_at
Expression level (log2)
Age
Smoker
non−smoker

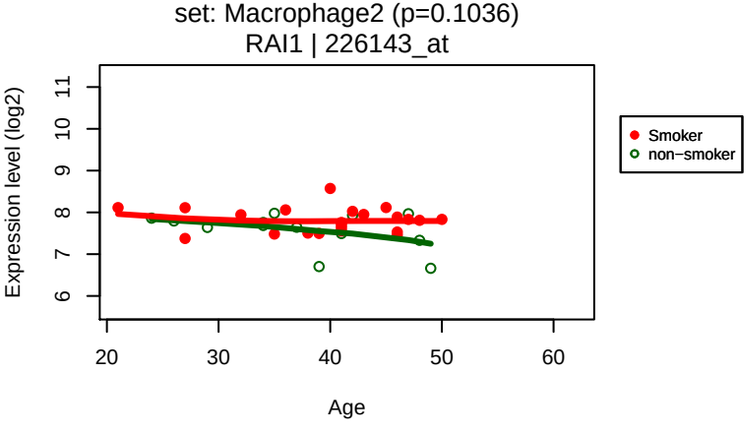

set: SmallAirway2 (p=0.5657)
RAI1 | 226143_at
Expression level (log2)
Age
Smoker
non−smoker

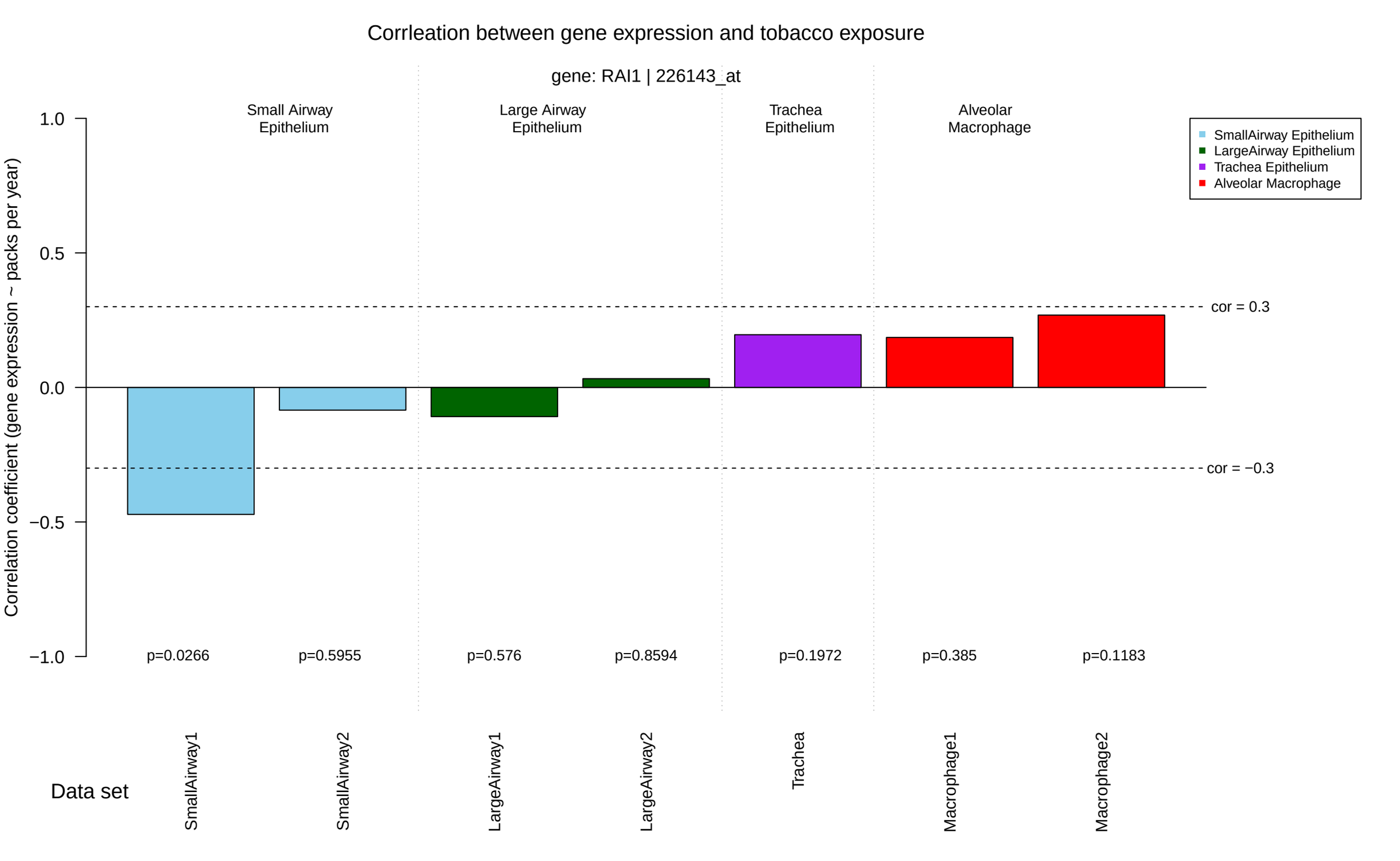

set: LargeAirway1 (p=0.8828)
RAI1 | 226143_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4756)
RAI1 | 226143_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0183)
RAI1 | 226143_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9366)
RAI1 | 226143_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1036)
RAI1 | 226143_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAI1 | 226143_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0266
p=0.5955
p=0.576
p=0.8594
p=0.1972
p=0.385
p=0.1183
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2